Brain Tumour MRI Image Classification

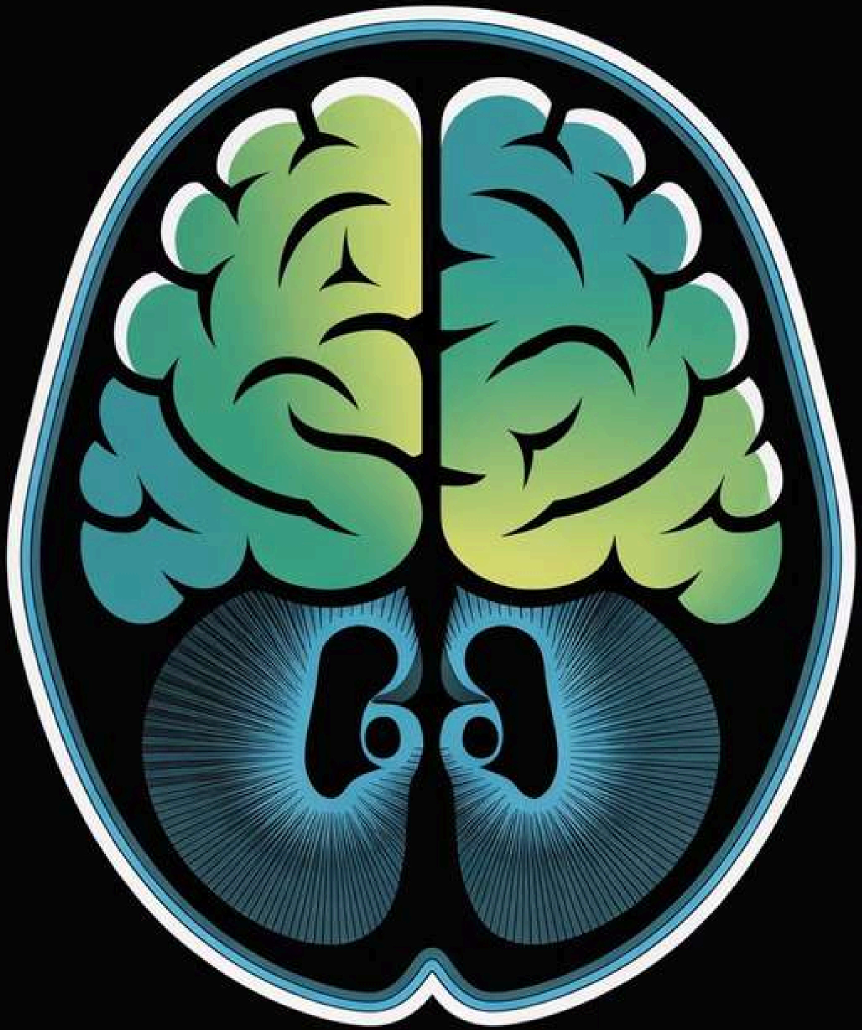
A deeplearning-based approach to detect and classify brain tumours from MRI scans using AI-powered medical imaging analysis.

Developed By: Yogapriya

Domain: Medical Imaging Deep Learning & AI

Technologies: Python, TensorFlow/Keras, Streamlit

Dataset: Brain MRI Images (4 Classes)



Problem Statement & Objective

The Challenge

Braintumour diagnosis through MRI images is time-consuming and prone to human error. Manual interpretation by radiologists can delay critical treatment decisions.

Our Solution

Developan AI-powered system that classifies MRI brain images into tumour types, supporting early diagnosis and faster decision-making to assist radiologists.

01

Build Custom CNN

Develop convolutional neural network architecture

02

Apply Transfer Learning

Leverage pre-trained models for classification

03

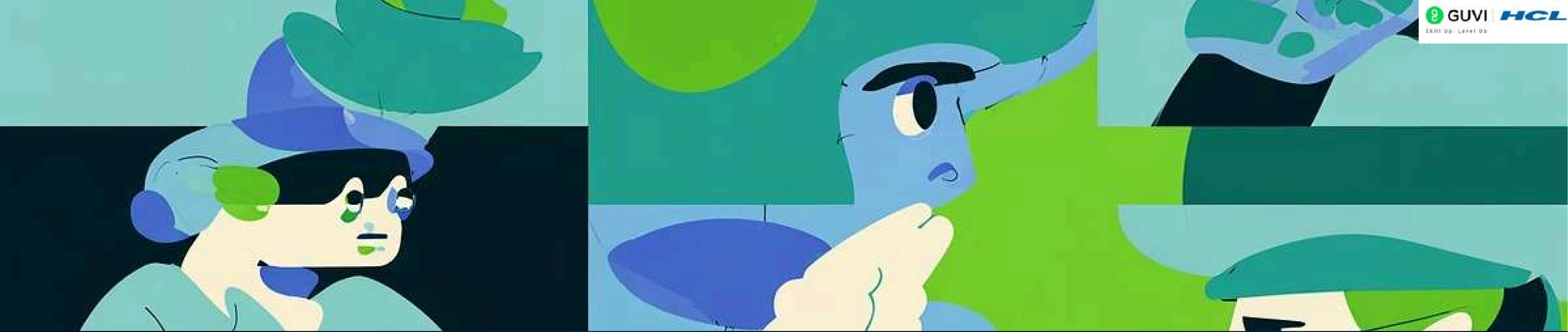
Compare Performance

Evaluate and select best-performing model

04

Deploy via Streamlit

Create accessible web application interface



Dataset Overview

Four Tumour Categories

- Glioma
- Meningioma
- Pituitary
- No Tumour

Dataset Characteristics

Source: Brain MRITumourDataset

Image Size: All images resized to 224×224 pixels

Balance: Equal distribution across all four classes

Normalisation

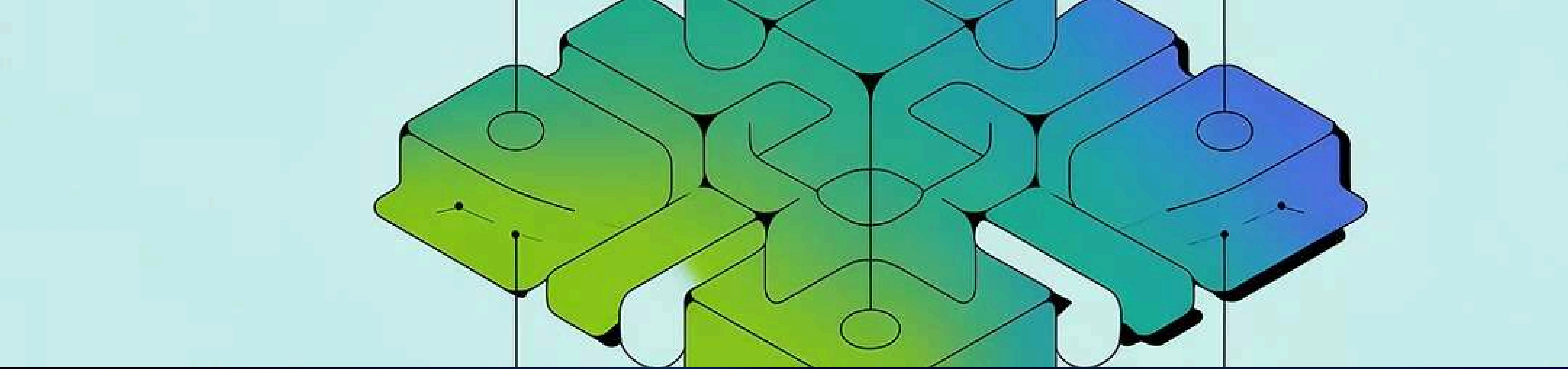
Pixel values scaled to 0-1 range for consistent model input

Image Resizing

Standardised dimensions ensure uniform processing across dataset

Data Augmentation

Rotation, flip,zoom,and brightness adjustments enhance model robustness



Model Building & Training

Custom CNN Model

- Convolution, Pooling, Dropout,Dense layers
- Batch Normalisation and ReLU activation
- Optimizer: Adam | Loss: Categorical Crossentropy

Transfer Learning Models

- Pre-trained: ResNet50,InceptionV3, MobileNetV2, EfficientNetB0
- Customised layers for 4-class classification
- Fine-tuned top layers for enhanced accuracy

Training Strategy

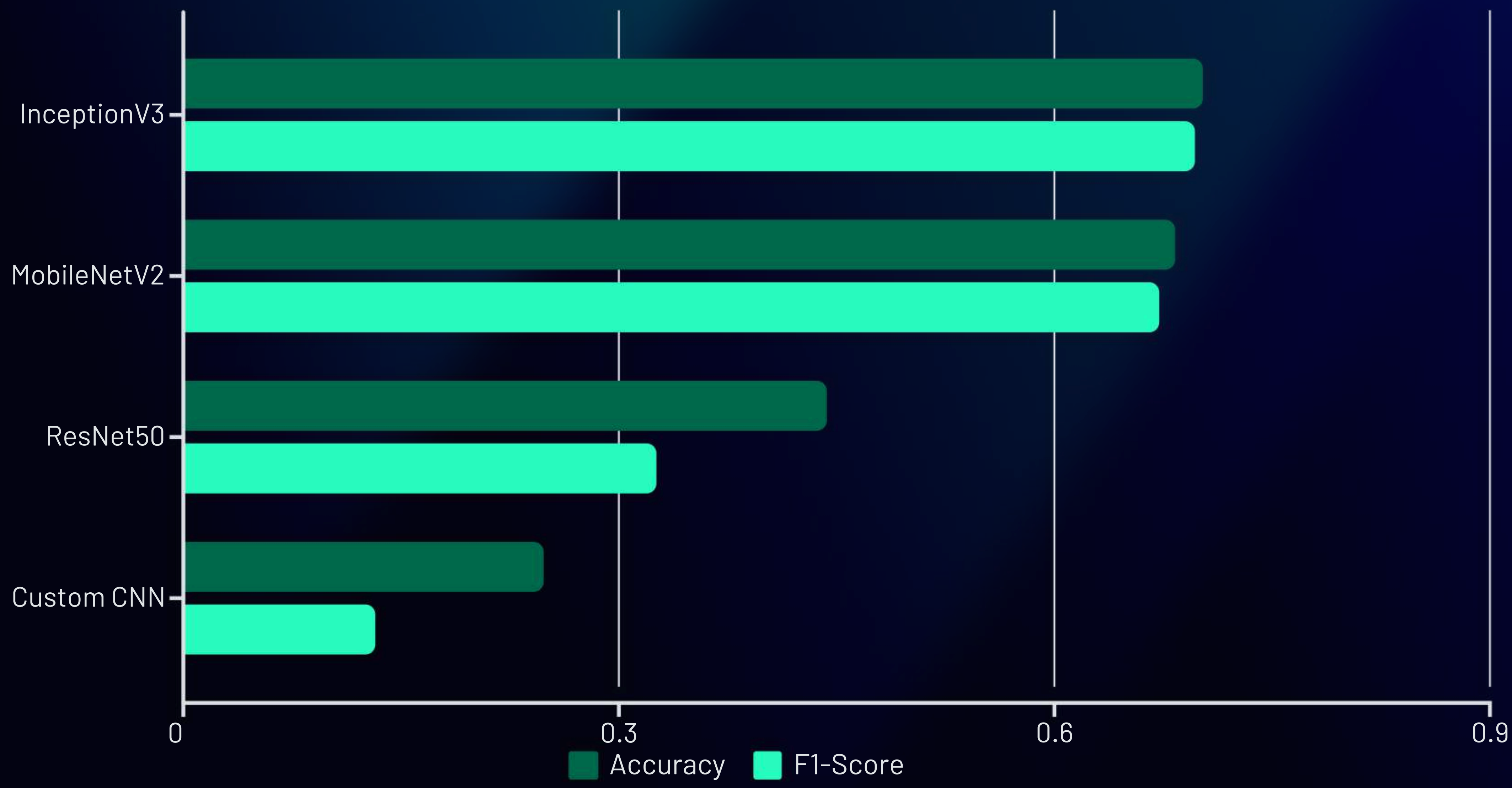
Models trained with EarlyStopping and ModelCheckpoint callbacks to monitor validation loss and prevent overfitting.

Architecture Highlights

Each model optimised for medical imaging tasks with balanced depth and computational efficiency for accurate tumour classification.

Model Evaluation

Comprehensive performance analysis across multiple deep learning architectures using standard classification metrics.



70.3%

BestAccuracy

InceptionV3 achieved highest performance

0.719

TopPrecision

InceptionV3 precision score

0.697

F1-Score

Balanced performance metric

Key Insight: InceptionV3 demonstrated the most balanced precision, recall, and F1-score, making it the optimal choice for deployment. Accuracy vs. loss curves plotted during training validated model convergence.

Streamlit Deployment

Upload

User uploads MRI scan image (.jpg/.png)

Preprocess

Image resized and normalised automatically

Predict

Modelclassifies tumour type with confidence

Show Results

Display predictions with confidence scores

App Features

- Upload MRI image for instant analysis
- Model predicts tumour type with confidence score
- Displays confidence bar chart for all four classes
- User-friendly, responsive interface with theme styling

Prediction Classes

- Glioma
- Meningioma
- Pituitary
- No Tumour

Platform: Streamlit App | **Title:** Brain Tumour Classifier | Simple, interactive, and lightweight UI with built-in "About" section

Conclusion & Future Scope

Key Achievement

Successfully classified brain MRI images into four tumour types using deep learning. InceptionV3 achieved 70.3% accuracy with the most balanced precision, recall, and F1-score.

Deployment Success

Streamlitwebappenables real-time MRI image uploads and displays instant tumour-type predictions with confidence scores for clinical support.

Current Limitations

Computational Constraints: Models trained using Google Colab CPU, restricting capacity and training speed.

Architecture Limits: Heavy architectures like EfficientNetB0 couldn't be fully trained due to compute limits.

Future Enhancements

GPU/TPU Training

Re-trainmodelsusing GPU or TPU to reduce training time and enhance accuracy significantly.

EfficientNetB0

ExploreEfficientNetB0 for higher accuracy and computational efficiency once GPU resources are available.

Tumour Segmentation

IntegrateGrad-CAM andsegmentation modules for visual tumour localisation along with classification.

Dataset Expansion

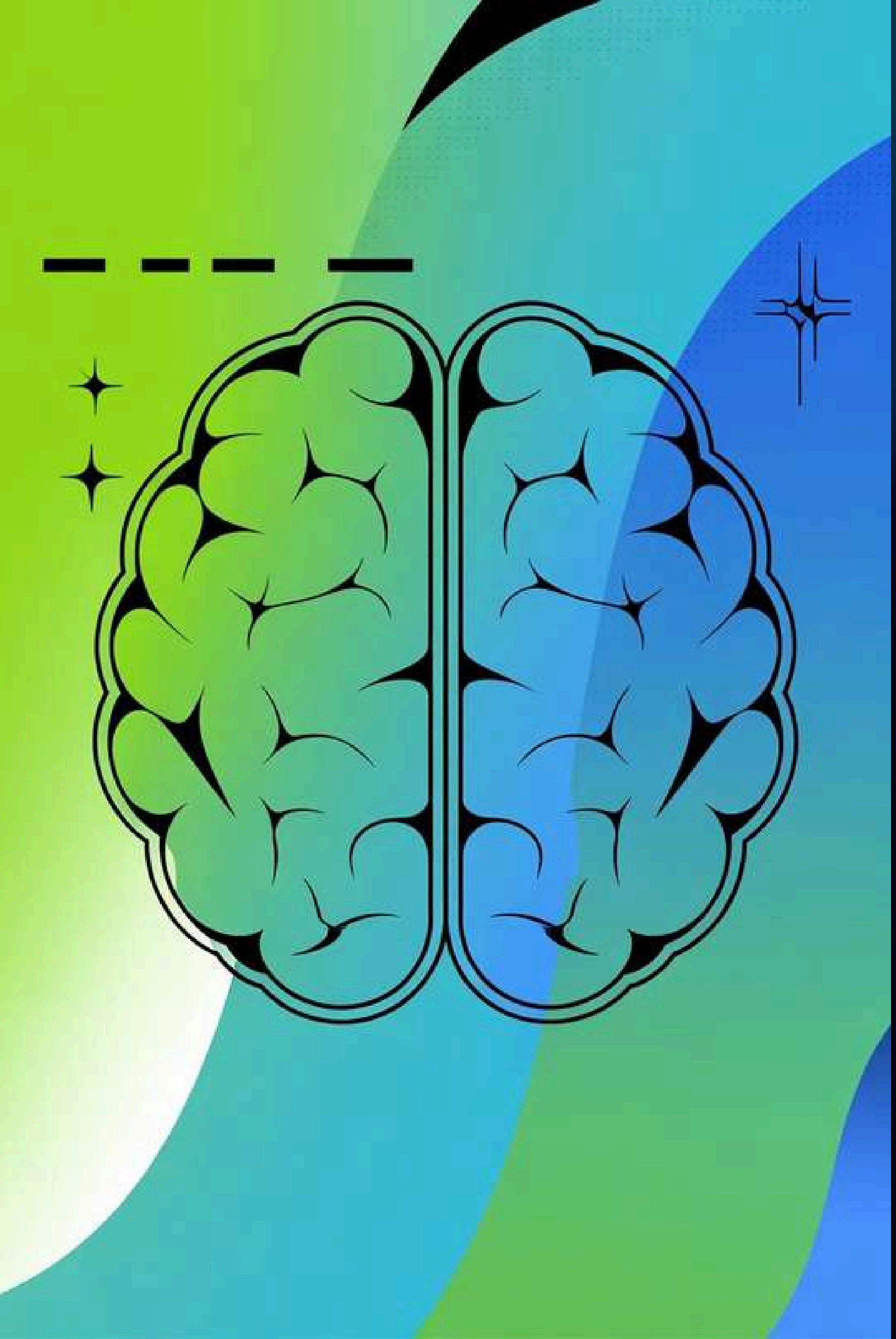
Expanddatasetsizeanddiversity for better model generalisation across varied patient populations.

Cloud Deployment

DeployonAWS/GCPforreal-time hospital or telemedicine use with scalable infrastructure.

Edge Computing

Explorelightweightedge-compatible models for mobile and point-of-care diagnostic applications.



Thank You!

Brain Tumour MRI Image Classification Project

Empowering medical diagnosis through artificial intelligence